

Five Zak Zeros for a Generalized First Hermite Function

A counterexample to Conjecture 4.1 of arXiv:2502.09510

fa_banach_001, lane 4

August 17, 2026

Abstract

Faulhuber conjectured that every dilated and chirped first Hermite function has exactly the three Zak-transform zeros forced by odd parity. We give an exact hexagonal counterexample. For

$$a = \sqrt{\frac{2}{\sqrt{3}}}, \quad s = \frac{1}{2},$$

the generalized first Hermite function has two additional Zak zeros at $(1/6, 5/6)$ and $(5/6, 1/6)$. The proof identifies the squared norm of a Gaussian Zak transform with an absolutely convergent theta series and uses its order-three affine symmetry. No numerical approximation enters the proof.

1 Source conjecture and result

For a suitable function f , use the convention

$$\mathcal{Z}f(x, \omega) = \sum_{k \in \mathbb{Z}} f(k - x) e^{2\pi i k \omega}. \quad (1)$$

The first Hermite function is a nonzero constant multiple of $te^{-\pi t^2}$. Section 4 of M. Faulhuber, *Gabor systems with Hermite functions of order n and oversampling greater than $n + 1$ which are not frames*, arXiv:2502.09510, makes the following conjecture. If

$$\tilde{h}_1(t; a, s) = \frac{1}{\sqrt{a}} h_1\left(\frac{t}{a}\right) e^{\pi i s t^2}, \quad a > 0, \quad s \in \mathbb{R}, \quad (2)$$

then $\mathcal{Z}\tilde{h}_1$ has exactly three zeros in $[0, 1)^2$, namely

$$(0, 0), \quad (1/2, 0), \quad (0, 1/2). \quad (3)$$

These three zeros are indeed always present because (2) is odd.

Theorem 1. *Conjecture 4.1 of arXiv:2502.09510 is false. Set*

$$a_0 = \sqrt{\frac{2}{\sqrt{3}}}, \quad s_0 = \frac{1}{2}. \quad (4)$$

In addition to the three zeros in (3), one has

$$\mathcal{Z}\tilde{h}_1\left(\frac{1}{6}, \frac{5}{6}; a_0, s_0\right) = 0, \quad \mathcal{Z}\tilde{h}_1\left(\frac{5}{6}, \frac{1}{6}; a_0, s_0\right) = 0. \quad (5)$$

Thus this generalized first Hermite function has at least five distinct Zak zeros in the fundamental square.

2 Gaussian reduction

Let

$$q = a^{-2} - is, \quad \tau = iq = s + \frac{i}{a^2}, \quad \phi_q(t) = e^{-\pi q t^2}, \quad (6)$$

so that $\Re q > 0$ and $\Im \tau > 0$. Put

$$\Theta(x, \omega) = \mathcal{Z}\phi_q(x, \omega), \quad F(x, \omega) = |\Theta(x, \omega)|^2. \quad (7)$$

All series below, together with their derivatives, converge normally.

Lemma 2. *At every point where $\Theta \neq 0$,*

$$\nabla F = 0 \quad \Longleftrightarrow \quad \Theta_x = 0. \quad (8)$$

Moreover, for a nonzero constant $C_{a,s}$,

$$\mathcal{Z}\tilde{h}_1(x, \omega; a, s) = C_{a,s} \Theta_x(x, \omega). \quad (9)$$

Proof. Since $h_1(t) = C_1 t e^{-\pi t^2}$ with $C_1 \neq 0$, (2) and (6) give

$$\tilde{h}_1(t; a, s) = C_1 a^{-3/2} t e^{-\pi q t^2}.$$

Termwise differentiation of (7) yields

$$\Theta_x = 2\pi q \sum_{k \in \mathbb{Z}} (k - x) e^{-\pi q (k-x)^2} e^{2\pi i k \omega},$$

which proves (9).

The same series gives the first-order identity

$$\Theta_\omega = \frac{i}{q} \Theta_x + 2\pi i x \Theta. \quad (10)$$

At a point where $\Theta \neq 0$, write $r = \Theta_x / \Theta$. The equation $F_x = 0$ says $\Re r = 0$, hence $r = it$ for some $t \in \mathbb{R}$. By (10),

$$\frac{F_\omega}{2F} = \Re \left(\frac{i}{q} r + 2\pi i x \right) = -t \Re \left(\frac{1}{q} \right).$$

Because $\Re q > 0$, one has $\Re(1/q) > 0$; therefore $F_\omega = 0$ if and only if $t = 0$, or equivalently $\Theta_x = 0$. The reverse implication follows from the same formulas. $\square$

3 The hexagonal affine symmetry

Write

$$\tau = c + ib, \quad b > 0,$$

so $q = b - ic$. The next Fourier expansion is the source of the exact symmetry.

Lemma 3. *The periodic function F has the absolutely convergent Fourier series*

$$F(x, \omega) = \frac{1}{\sqrt{2b}} \sum_{n, m \in \mathbb{Z}} (-1)^{mn} \exp\left(-\frac{\pi}{2b} |n + m\tau|^2\right) e^{2\pi i(n\omega + m\omega)}. \quad (11)$$

Proof. Let $\widehat{F}(n, m)$ denote the Fourier coefficient obtained by multiplying by $e^{-2\pi i(n\omega + m\omega)}$ and integrating over the unit square. Expanding $|\Theta|^2$, integration in ω imposes a difference of indices equal to m . Joining the remaining unit intervals then gives

$$\widehat{F}(n, m) = \int_{\mathbb{R}} \phi_q(t + m) \overline{\phi_q(t)} e^{2\pi i n t} dt.$$

Completing the square in this Gaussian integral yields

$$\widehat{F}(n, m) = \frac{1}{\sqrt{2b}} \exp\left(-\frac{\pi}{2b} |n + m\tau|^2 - \pi i m n\right),$$

which is (11). Gaussian decay also proves absolute convergence. $\square$

For the parameters (4),

$$\tau = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad b = \frac{\sqrt{3}}{2}, \quad \tau^2 = \tau - 1. \quad (12)$$

Define the integer matrix and its inverse transpose by

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \quad B = A^{-T} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}. \quad (13)$$

The map A sends (n, m) to $(-m, n + m)$, and (12) gives

$$\tau(n + m\tau) = -m + (n + m)\tau. \quad (14)$$

It therefore preserves the Gaussian factor in (11). Its sign changes by

$$(-1)^{(-m)(n+m)} = (-1)^m (-1)^{mn}. \quad (15)$$

Reindexing (11) with (14)–(15) proves

$$F(S(x, \omega)) = F(x, \omega), \quad S(x, \omega) = \left(x - \omega + \frac{1}{2}, x\right) \pmod{\mathbb{Z}^2}. \quad (16)$$

Lemma 4. *The points*

$$p_1 = \left(\frac{1}{6}, \frac{5}{6}\right), \quad p_0 = \left(\frac{1}{2}, \frac{1}{2}\right), \quad p_2 = \left(\frac{5}{6}, \frac{1}{6}\right) \quad (17)$$

are critical points of F .

Proof. From (13) and (16),

$$S^2(x, \omega) = \left(-\omega, x - \omega + \frac{1}{2} \right) \pmod{\mathbb{Z}^2}. \quad (18)$$

Solving $S^2(x, \omega) = (x, \omega)$ on the torus gives exactly the three points (17). The derivative of S^2 is

$$B^2 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

whose eigenvalues are the primitive cube roots of unity; in particular it has no eigenvalue 1. At a fixed point p , differentiating $F \circ S^2 = F$ gives

$$(B^2)^T \nabla F(p) = \nabla F(p).$$

Thus $\nabla F(p) = 0$. □

4 Proof of the counterexample

It remains only to exclude the possibility that the critical points p_1, p_2 are zeros of Θ , because Lemma 2 was stated away from those zeros. With the standard Jacobi theta function

$$\vartheta_3(z; \tau) = \sum_{k \in \mathbb{Z}} e^{\pi i \tau k^2 + 2\pi i k z},$$

expansion of the square in (7) gives

$$\Theta(x, \omega) = e^{\pi i \tau x^2} \vartheta_3(\omega - \tau x; \tau). \quad (19)$$

The classical theta function $\vartheta_3(\cdot; \tau)$ has one simple zero modulo $\mathbb{Z} + \tau\mathbb{Z}$, at $(1 + \tau)/2$. Consequently Θ has exactly one zero modulo $\mathbb{Z}^2$, namely

$$(x, \omega) = \left(\frac{1}{2}, \frac{1}{2} \right) = p_0. \quad (20)$$

For completeness, the theta zero count follows immediately by integrating ϑ'_3/ϑ_3 around a period parallelogram and using the standard quasi-periodicity; the displayed half-period is a zero by pairing terms.

It follows from (20) that $\Theta(p_1)$ and $\Theta(p_2)$ are nonzero. Lemma 4 and Lemma 2 now give

$$\Theta_x(p_1) = \Theta_x(p_2) = 0.$$

Equation (9) proves the two identities (5). Together with the three parity-forced zeros (3), this completes the proof of Theorem 1.

5 Scope, verification, and novelty

The theorem is a complete negative answer to the paper's explicit Conjecture 4.1. It does not determine the full zero set, nor does it answer Questions Q1–Q3 for every Hermite order. Applying

the source paper’s periodic-zero criterion to the five displayed zeros also gives a density-five periodic non-frame Gabor system for this generalized first Hermite window.

The accompanying high-precision script evaluates all five zeros and checks the affine norm symmetry at sample points. It is only a regression test; the proof above is exact.

A bounded novelty search covered the run indexes, the exact conjecture, the two rational zero locations, generalized first Hermite functions, hexagonal lattices, and related theta/Zak terminology. It also inspected the closely related arXiv:2502.08358, which uses a theta identity for a second-Hermite example. No prior statement of (5) or answer to Conjecture 4.1 was found. Long–Wei–Zou’s 2025 classification of critical points of heat kernels on tori reports the closely analogous six-critical-point phenomenon on the hexagonal torus, providing conceptual corroboration but not, in the bounded search, this Zak/Hermite specialization. Novelty is therefore provisional pending expert bibliographic review.

References

- [1] M. Faulhuber, *Gabor systems with Hermite functions of order n and oversampling greater than $n + 1$ which are not frames*, Sampling Theory, Signal Processing, and Data Analysis 23 (2025); arXiv:2502.09510.
- [2] M. Faulhuber, *Operator-isomorphism pairs and Zak transform methods for the study of Gabor systems*, arXiv:2502.08358, 2025.
- [3] C.-G. Long, J. Wei, and W. Zou, *The classification of critical points of heat kernel on tori and application to elliptic equation and Mueller–Ho conjecture*, Ann. Sc. Norm. Super. Pisa Cl. Sci., 2025.